Modeling and Passivity-Based Current Control of a 2 MW Direct-Drive PMSG Wind Energy Conversion System

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***Abstract*—This paper presents the modeling of a 2 MW direct-drive wind energy conversion system based on a permanent magnet synchronous generator (PMSG), including the turbine aerodynamics, the generator, the back-to-back converter, the DC link, the L filter and the grid, and the design of a passivity-based current controller for the machine-side converter. The generator is described in the receptor convention with parameters taken from published data, and the consistency of all the signs of the model is verified by an exact power balance between the aerodynamic power, the stored energy, the losses and the power injected into the grid. A step and a turbulent wind profile are defined for the evaluation of the system. The passivity-based controller keeps the workless coupling of the machine model and injects damping; its stability is proven with a storage function. Simulations with a sampling period equal to half of the switching period and a one-sample computation delay give a rise time of 3 ms without overshoot, a cross-coupling below 0.5 % of the reference step and a zero steady-state error.**

***Index Terms*—PMSG, wind energy conversion, modeling, passivity-based control, back-to-back converter.**

I. Introduction

Wind energy is one of the main sources in the transition towards low-carbon electricity generation [1], [2]. Direct-drive PMSG turbines remove the gearbox, which reduces maintenance and losses, and their full-scale back-to-back converter decouples the generator from the grid [3], [4]. The machine-side converter (MSC) controls the generator torque to track the maximum power point, and the grid-side converter (GSC) regulates the DC-link voltage and injects the power into the grid.

The machine-side currents are commonly controlled by PI loops in rotating frames [5], [6]. In this paper, passivity-based control (PBC) [7] is used for these currents, since it keeps the physical structure of the machine model.

The contributions of this paper are:

- a complete model of a 2 MW direct-drive PMSG wind energy conversion system, with generator, converter and grid data taken from published work, and a verification of its signs by an exact power balance;
- a step and a turbulent wind profile for the evaluation of the system;
- a passivity-based current controller for the MSC, with a stability proof and a discrete-time validation including the sampling and the computation delay.

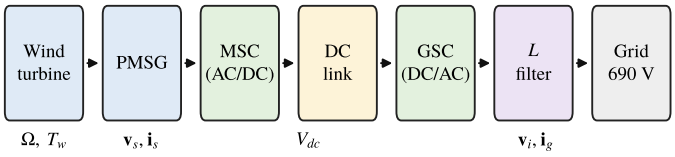


Fig. 1. Direct-drive PMSG wind energy conversion system with a back-to-back converter.

II. System Description and Turbine Model

The studied system is shown in Fig. 1. The turbine drives the PMSG directly; the MSC rectifies the generator voltages, the DC link decouples the two converters, and the GSC injects the power into the 690 V, 50 Hz grid through an L filter.

The power captured by the rotor and the tip-speed ratio are [8], [9]

$$P_w = \tfrac{1}{2}\rho\pi R^2 C_p(\lambda,\beta)\, v^3, \qquad \lambda = \frac{\Omega R}{v} \tag{1}$$

where ρ is the air density, R the rotor radius, v the wind speed, Ω the rotor speed and β the pitch angle. The power coefficient is

$$C_p = 0.5176\Big(\frac{116}{\lambda_i} - 0.4\beta - 5\Big)e^{-21/\lambda_i} + 0.0068\lambda, \tag{2}$$

with $1/\lambda_i = 1/(\lambda + 0.08\beta) - 0.035/(\beta^3+1)$. For $\beta = 0$, its maximum $C_{p,\max} = 0.48$ is reached at $\lambda_{opt} = 8.1$ (Fig. 2). The aerodynamic torque is $T_w = P_w/\Omega$.

For a rated power $P_n = 2$ MW and $R = 40$ m, close to the 2 MW turbine of [11] ($R = 38$ m), the rated wind speed follows from $P_n = \frac{1}{2}\rho\pi R^2 C_{p,\max} v_n^3$, which gives $v_n = 11.0$ m/s, and the rated rotor speed from $\Omega_n = \lambda_{opt} v_n / R = 2.23$ rad/s. The rated torque is $T_n = P_n/\Omega_n = 0.897$ MN·m.

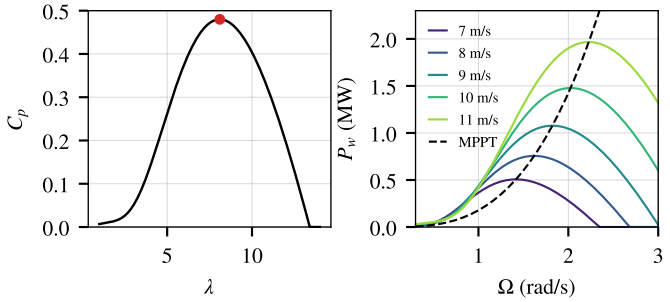


Fig. 2. Power coefficient for $\beta = 0$ (left) and captured power versus rotor speed for several wind speeds, with the maximum power curve $P_{opt} = K_{opt}\Omega^3$ (right).

The maximum power curve is $P_{opt} = K_{opt}\Omega^3$ with $K_{opt} = \frac{1}{2}\rho\pi R^5 C_{p,\max}/\lambda_{opt}^3$.

The direct-drive shaft is described by the one-mass model

$$J\frac{d\Omega}{dt} = T_w + T_e - B\Omega \tag{3}$$

where J is the total inertia, B the friction coefficient and T_e the electromagnetic torque in the receptor convention ($T_e < 0$ in generating mode).

III. Wind Speed Model

Two wind profiles are used (Fig. 3). The step profile (9, 11 and 8 m/s), similar to the one of [9], is used to assess the transient response. The turbulent profile combines a mean value, low-frequency harmonics [12] and a turbulence:

$$v(t) = V_m + \textstyle\sum_{k=1}^{3} A_k \sin(2\pi f_k t) + v_t(t) \tag{4}$$

with $V_m = 8.5$ m/s, $A = [1.5,\ 0.2,\ 0.3]$ m/s and $f = [0.03,\ 0.16,\ 0.17]$ Hz. The turbulence v_t is a first-order coloured noise with a standard deviation of 0.5 m/s and a correlation time of 1.5 s, generated by a Park–Miller generator and the Box–Muller transform, so that the same sequence is obtained on any platform. Over 40 s, the wind has a mean value of 8.6 m/s, a standard deviation of 1.1 m/s and remains between 6.0 and 10.9 m/s, i.e., below the rated wind speed, in the maximum power point tracking region where the pitch angle is zero.

IV. PMSG Model

In the rotor dq frame (amplitude-invariant transformation, d axis aligned with the magnet flux), with the receptor convention and $L_d = L_q = L_s$, the PMSG is described by [9]

$$v_{sd} = R_s i_{sd} + L_s\frac{di_{sd}}{dt} - \omega_e L_s i_{sq} \tag{5}$$

$$v_{sq} = R_s i_{sq} + L_s\frac{di_{sq}}{dt} + \omega_e L_s i_{sd} + \omega_e\psi_f \tag{6}$$

$$T_e = \tfrac{3}{2} p\,\psi_f\, i_{sq} \tag{7}$$

where $\omega_e = p\Omega$, p is the number of pole pairs and ψ_f the magnet flux; in generating mode, $i_{sq} < 0$. With $\mathbf{i}_s = [i_{sd}\ i_{sq}]^T$, $\mathbf{v}_s = [v_{sd}\ v_{sq}]^T$, $\mathbf{e}_q = [0\ 1]^T$ and the skew-symmetric matrix $\boldsymbol{\mathcal{J}} = \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$,

$$L_s\dot{\mathbf{i}}_s = \mathbf{v}_s - R_s\mathbf{i}_s + \omega_e L_s\boldsymbol{\mathcal{J}}\mathbf{i}_s - \omega_e\psi_f\mathbf{e}_q. \tag{8}$$

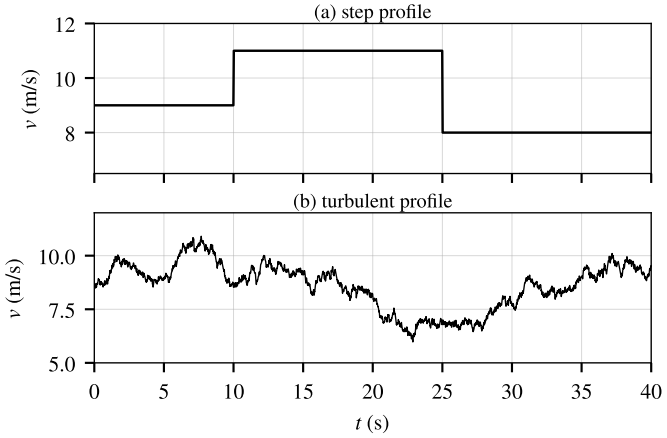


Fig. 3. Wind speed profiles: (a) steps of 9, 11 and 8 m/s; (b) turbulent profile (4).

TABLE I
Parameters of the Studied System

Parameter	Symbol	Value	Source
Rated power	P_n	2 MW	[10]
Rotor radius	R	40 m	[11]
Air density	ρ	1.225 kg/m^3	[9]
Rated wind / rotor speed	v_n, Ω_n	11.0 m/s, 2.23 rad/s	(1)
Rated torque	T_n	0.897 MN·m	P_n/Ω_n
Total inertia	J	2.8×10^6 kg·m^2	assumed
Friction coefficient	B	10^3 N·m·s/rad	assumed
Pole pairs	p	26	[10]
Stator resistance	R_s	0.78 mΩ	[10]
Stator inductance	L_s	1.57 mH	[10]
Magnet flux	ψ_f	9.18 Wb	[10]
DC-link capacitance	C	20 mF	[10]
DC-link voltage	V_{dc}	1200 V	[10]
Filter	L_f, R_f	0.15 mH, 2 mΩ	[10], assumed
Grid		690 V, 50 Hz	[10]
Switching frequency	f_{sw}	2160 Hz	[10]

The generator data are those of the non-salient 2 MW PMSG of [10] (Table I). At Ω_n, the back-EMF $p\Omega_n\psi_f = 532$ V is close to the peak phase voltage of the 690 V grid (563 V), and the rated torque requires $i_{sq} = -T_n/(\frac{3}{2}p\psi_f) = -2.51$ kA, consistent with the rated current of about 2.6 kA given in [10].

The model is first verified with the generator driven at Ω_n and connected to a balanced resistive load of 0.19 Ω per phase ($\mathbf{v}_s = -R_L\mathbf{i}_s$). Fig. 4 shows that the currents settle in about 40 ms to $i_{sd} = -1.08$ kA and $i_{sq} = -2.27$ kA. The electromagnetic torque is negative (-0.814 MN·m), i.e., it opposes the rotation, and the phase currents are balanced sinusoids at the electrical frequency of 9.2 Hz. In steady state, the mechanical power absorbed by the generator, 1.8144 MW, equals the power delivered to the load, 1.8069 MW, plus the copper losses, 7.4 kW, with a residual below 10^{-7} W.

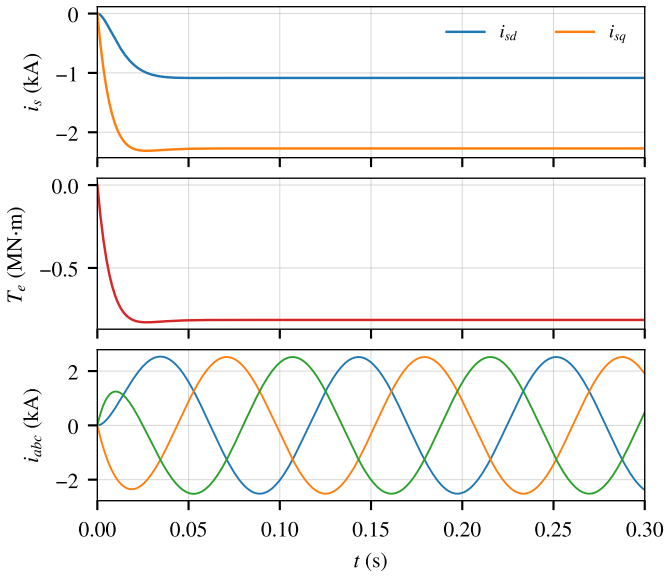


Fig. 4. PMSG driven at Ω_n and connected to a resistive load: dq currents, electromagnetic torque and phase currents i_a, i_b, i_c.

V. Converter, DC Link and Grid Model

The two-level converters are represented by their average model; the voltage vectors are limited to $V_{dc}/\sqrt{3}$ (space-vector modulation). The DC link receives the power of the MSC and supplies the GSC [9]:

$$CV_{dc}\frac{dV_{dc}}{dt} = P_{dc} - P_i, \quad P_{dc} = -\tfrac{3}{2}\mathbf{v}_s^T\mathbf{i}_s, \quad P_i = \tfrac{3}{2}\mathbf{v}_i^T\mathbf{i}_g. \tag{9}$$

In the grid-voltage-oriented frame ($v_{gq} = 0$), the L filter between the GSC voltage $\mathbf{v}_i$ and the grid voltage $\mathbf{v}_g$, with the current $\mathbf{i}_g$ counted towards the grid, gives [9]

$$L_f\dot{\mathbf{i}}_g = \mathbf{v}_i - R_f\mathbf{i}_g + \omega_g L_f\boldsymbol{\mathcal{J}}\mathbf{i}_g - \mathbf{v}_g, \tag{10}$$

and the grid powers are $P_g = \frac{3}{2}v_{gd}i_{gd}$ and $Q_g = -\frac{3}{2}v_{gd}i_{gq}$.

Equations (3), (8)–(10) form a sixth-order model with state $[i_{sd}, i_{sq}, \Omega, V_{dc}, i_{gd}, i_{gq}]$. With the stored energy $E = \frac{1}{2}J\Omega^2 + \frac{3}{4}L_s\|\mathbf{i}_s\|^2 + \frac{1}{2}CV_{dc}^2 + \frac{3}{4}L_f\|\mathbf{i}_g\|^2$, and since $\mathbf{x}^T\boldsymbol{\mathcal{J}}\mathbf{x} = 0$ and $T_e\Omega = \frac{3}{2}\omega_e\psi_f i_{sq}$, the model gives, for any state and any converter voltages,

$$\dot{E} = P_w - B\Omega^2 - \tfrac{3}{2}R_s\|\mathbf{i}_s\|^2 - \tfrac{3}{2}R_f\|\mathbf{i}_g\|^2 - P_g. \tag{11}$$

The aerodynamic power is thus stored, dissipated or delivered to the grid; this identity checks the sign of every term of the model.

Table II gives the operating point at $v = 11$ m/s, with Ω at its optimal value, $i_{sd} = 0$ and unity power factor. The converter voltages remain below the limit $V_{dc}/\sqrt{3} = 693$ V. Simulated during 1 s with constant converter voltages, the model stays at this equilibrium, and the power flow satisfies (11) with a residual below 10^{-8} W.

TABLE II
Operating Point at $v = 11$ m/s

Quantity	Symbol	Value
Rotor speed	Ω	2.228 rad/s
Stator / grid currents	i_{sq}, i_{gd}	-2460 A, 2295 A
MSC / GSC voltages	$\|\mathbf{v}_s\|$, $\|\mathbf{v}_i\|$	575 V, 578 V
Aerodynamic power	P_w	1.967 MW
Friction / stator / filter losses		5.0 kW, 7.1 kW, 15.8 kW
Grid power	P_g	1.939 MW

TABLE III
Step Response of the Current Loop

Indicator	Value
Rise time (10–90 %), i_{sq} step of -1 kA	3.0 ms
Overshoot	0 %
Maximum $\|i_{sd}\|$ during the i_{sq} steps	10.7 A
Steady-state error	0 A

VI. Passivity-Based Current Control of the MSC

With $\mathbf{e}_s = \mathbf{i}_s - \mathbf{i}_s^*$ and $\dot{\mathbf{z}}_s = \mathbf{e}_s$, the MSC voltage reference is

$$\mathbf{v}_s^* = L_s\dot{\mathbf{i}}_s^* + R_s\mathbf{i}_s^* - \omega_e L_s\boldsymbol{\mathcal{J}}\mathbf{i}_s^* + \omega_e\psi_f\mathbf{e}_q - R_a\mathbf{e}_s - K_i\mathbf{z}_s. \tag{12}$$

The first four terms reproduce the desired dynamics of (8) along the reference, with the coupling evaluated on the reference so that it remains workless; R_a injects damping and K_i removes the steady-state error. The reference derivative is obtained by a first-order filter (2 ms), and the integral is frozen when $\|\mathbf{v}_s^*\|$ reaches $V_{dc}/\sqrt{3}$.

Proposition 1. *With (12) applied to (8) and $R_a, K_i > 0$, the storage function $H = \frac{1}{2}L_s\|\mathbf{e}_s\|^2 + \frac{1}{2}K_i\|\mathbf{z}_s\|^2$ satisfies $\dot{H} = -(R_s + R_a)\|\mathbf{e}_s\|^2$ and $\mathbf{e}_s \to 0$.*

Proof. Subtracting $L_s\dot{\mathbf{i}}_s^*$ from (8) and inserting (12) gives $L_s\dot{\mathbf{e}}_s = -(R_s + R_a)\mathbf{e}_s + \omega_e L_s\boldsymbol{\mathcal{J}}\mathbf{e}_s - K_i\mathbf{z}_s$. Hence $\dot{H} = \mathbf{e}_s^T L_s\dot{\mathbf{e}}_s + K_i\mathbf{z}_s^T\mathbf{e}_s = -(R_s + R_a)\|\mathbf{e}_s\|^2$, since $\mathbf{e}_s^T\boldsymbol{\mathcal{J}}\mathbf{e}_s = 0$. H is bounded and $\dot{H}$ is uniformly continuous, so Barbalat's lemma [13] gives $\mathbf{e}_s \to 0$. □

The gains place a double closed-loop pole at $\omega_i = 500$ rad/s: $R_s + R_a = 2\omega_i L_s$ and $K_i = \omega_i^2 L_s$, i.e., $R_a = 1.569\ \Omega$ and $K_i = 392.5\ \Omega$/s.

The controller is tested with the rotor speed held at Ω_n. It is sampled at $T_s = 1/(2f_{sw}) = 231.5\ \mu$s (regular sampling with double update), the voltage computed at one sample is applied at the next one, and the plant is integrated with a fourth-order Runge–Kutta scheme. The references are steps of i_{sq}^* from 0 to -1 kA at 20 ms and to -2.5 kA (rated value) at 70 ms, and of i_{sd}^* from 0 to -0.5 kA at 120 ms (Fig. 5).

Table III summarizes the response. The currents follow their references with a rise time of 3 ms and no overshoot. The cross-coupling on i_{sd} during the i_{sq} steps is 10.7 A, i.e., about 0.4 % of the rated current, because the coupling terms are compensated in (12). The voltage reaches the limit $V_{dc}/\sqrt{3}$ only during the step of -1.5 kA, where the reference derivative is large; between the steps, $\|\mathbf{v}_s\|$ is about 575 V.

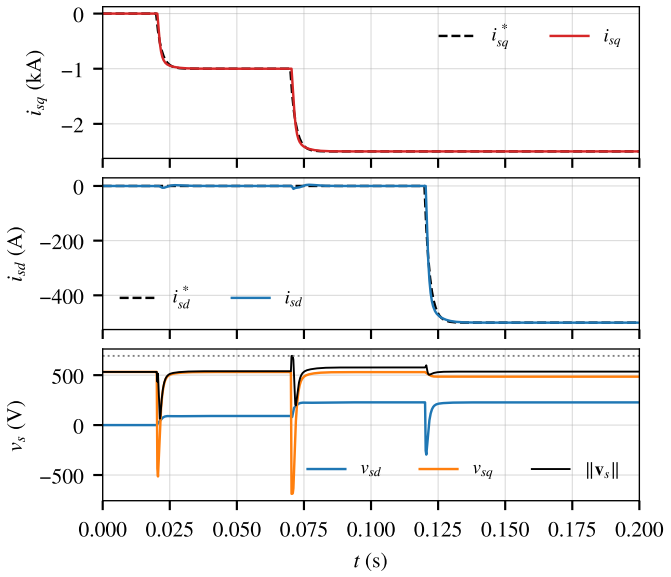


Fig. 5. Passivity-based current control of the MSC at $\Omega = \Omega_n$: q- and d-axis currents and their references, and stator voltages (dotted: limit $V_{dc}/\sqrt{3}$).

VII. Conclusion

A 2 MW direct-drive PMSG wind energy conversion system was modeled from the turbine to the grid, with generator, converter and grid data taken from published work and a turbine sized for the rated power. The model is written in the receptor convention, and an exact power balance between the aerodynamic power, the stored energy, the losses and the grid power verifies all its signs, both analytically and numerically. Step and turbulent wind profiles were defined for the evaluation. A passivity-based current controller for the machine-side converter, whose stability follows from a storage function, tracks the current references with a rise time of 3 ms, no overshoot and a negligible cross-coupling, with the sampling and the computation delay of a digital implementation.

Conflict of Interest

The authors declare there are no conflicts of interest.

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